

Algebraic Convergence Loci for Nonlinear Hahn–Fuchsian Systems

Finite resonances, bounded-exponent obstructions,
and universality with accumulating supports

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Abstract

Consider $Dy = Ay + F(x, y)$, where $D = x d/dx$, A is an arbitrary complex matrix, $F(0, 0) = 0$, $F_y(0, 0) = 0$, and the coefficients of F are absolutely convergent Hahn series supported in a fixed well-ordered positive real exponent monoid. Finite accumulation of exponents is allowed.

We prove that the positive, logarithm-free formal solutions are parametrized by a finite affine algebraic set of resonant coefficients. More importantly, *the subset of parameters producing absolutely convergent solutions is itself affine algebraic*, even when the spectral separation is zero. The proof separates a bounded exponent window from a contractive analytic tail, then takes a finite-dimensional algebraic quotient of coefficient families by ℓ^1 . The additional convergence equations have strictly bounded weighted degree. Convergent solutions have a common positive radius on each compact subset of their algebraic parameter locus.

This conclusion is sharp in a geometric sense: every affine algebraic set occurs, after a harmless normalization, as the convergence locus of a triangular system with *no formal compatibility obstruction*. We also prove an exact universal spectral-gap criterion, a finite-ancestry bound for formal obstructions, and a sharp nonlinear example in which the forcing stays away from the spectrum but quadratic interactions create an accumulation of small divisors. Its convergence threshold is exactly $p > 2$.

All theorem-level arguments are given in the article. The accompanying program checks 670 exact scalar equalities, including 21 finite systems. These computations are consistency checks, not a formal verification of the infinite-support results.

Status. A self-contained research extension of selected ProveIt material. Historical priority has not been independently established; no claim of a solution to a famous general transseries conjecture is made.

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1 The question and the proposed advance

Formal solvability, the absence of logarithms, and analytic convergence are three different questions. For ordinary regular-singular equations their relationship is familiar. With Hahn exponents, infinitely many terms can lie below one fixed spectral value, and formal division can destroy absolute convergence without changing the support.

The ProveIt package *Finite Resonance Certificates and Sharp Analytic Normalization for Hahn–Fuchsian Systems* treats a linear matrix-gauge problem. It identifies a universal separation condition involving spectral differences, and gives accumulation examples where a formal normalizer diverges [1]. Its further directions ask about constrained supports, cancellation, and the geometry of exceptional parameter sets. Those results are not reproved here as a new discovery. We change the problem: the unknown is a nonlinear solution germ, not a linear gauge, and we classify *which resonant parameters yield convergent germs*.

The precise research question is:

Can infinitely many small-divisor conditions impose a genuinely nonalgebraic convergence condition on the finite resonant parameters of a positive Hahn solution of a nonlinear regular-singular equation?

Within the class specified in Section 2, the answer is *no*. The convergence locus is algebraic. Conversely, it can be any affine algebraic set, so there is no general smoothness, linearity, or irreducibility theorem to replace this answer.

Editorial note (ProveIt, 2026-09-29). A package of the same batch, written independently, `Nonlinear_Hahn_Dulac_Finite_Resonance_Control/`, proves the universal criterion of Theorem 6.1 for real spectrum, with resonant constants and logarithms allowed in the solution (its `thm:universal`): one criterion, two independent proofs, which agree where both apply. Its accumulation example (`thm:sharp-family`, threshold $p > 2$) has forcing that itself accumulates at the eigenvalue, whereas in Theorem 8.1 the forcing is separated and the nonlinearity creates the accumulation. The linear counterpart of Theorem 8.1, safe edges whose composed path accumulates, is Example `ex:hidden` of `Path_Sensitive_Small_Divisors_Hahn_Fuchsian/`, again with threshold $p > 2$. Among the further directions of [1], the closest to the present question is “Geometry of the logarithmic strata” (the algebraic loci on which logarithms disappear); the convergence loci here are a nonlinear counterpart of those strata, not an answer to that question.

1.1 A guide to the results

There are two finite-dimensional obstructions, with very different meanings. The formal compatibility map Ψ depends on finitely many input coefficients. The analytic map χ has finitely many polynomial components, but can depend on the summability of infinite coefficient families. The main description is

$$\mathcal{V}_{\text{formal}} = \{c : \Psi(c) = 0\}, \quad \mathcal{V}_{\text{an}} = \{c : \Psi(c) = 0, \chi(c) = 0\}. \quad (1.1)$$

The second equality is not a finite-coefficient decision algorithm.

Theorem 3.1 constructs Ψ without an analytic assumption. Theorem 4.2 locates every possible convergence obstruction in one bounded exponent window. Theorem 5.1 converts that infinite window into the finite polynomial map χ . Theorem 7.1 realizes arbitrary affine algebraic sets as $\mathcal{V}_{\text{an}}$ while $\mathcal{V}_{\text{formal}}$ is affine space. Theorem 6.1 characterizes when χ vanishes for every input. The example in Section 8 shows why checking only the original forcing support misses nonlinear small divisors.

1.2 Relation to established theory

Generalized-series fixed-point methods are established. Van der Hoeven’s operator theory includes an implicit-function theorem for Noetherian operators [3]. We use a direct positive-valuation argument in the restricted setting needed here. Neumann support finiteness is also classical; the inspected repository already has a Lean development of finite ordered-word representations [2].

Convergence of generalized series satisfying analytic differential equations has substantial antecedents. Gontsov and Goryuchkina study complex-exponent series with real parts tending to infinity, and prove a majorant convergence theorem [4]; their later work develops a broader functional-equation framework [5]. Our accumulating coefficient supports are not an ordinary holomorphic power series in x at zero and are not covered merely by identifying our equation as regular singular. Absolute convergence for Hahn series itself is a standard notion, also used in higher-rank tropical geometry [6].

The proposed advance is the algebraicity and universality of the *parameter-dependent absolute-convergence locus at zero spectral gap*. The proof is supplied independently below. A targeted literature and repository comparison did not establish historical priority for this exact statement; the article does not assert exhaustive coverage of either.

1.3 Scope

Exponents are real and nonnegative, the exponent monoid is well ordered, and solutions have positive valuation. The constant matrix need not be semisimple and need not have real spectrum. We consider no powers of $\log x$ in the solution ansatz. Failure of formal compatibility means failure in this logarithm-free class, not nonexistence of other solutions. Divergence means failure of absolute Hahn convergence, not nonexistence of an analytic function with some weaker asymptotic interpretation.

Under $x = e^{-t}$, the monomials become $e^{-\gamma t}$ and $D = -d/dt$. Thus this is also a theorem about exponential transseries sectors with possibly accumulating positive actions. It is not a theorem about every logarithmic or exponential level of the full transseries field.

2 The coefficient algebra and the nonlinear model

2.1 A fixed well-ordered monoid

Let $\Gamma \subset \mathbb{R}_{\geq 0}$ be a nontrivial additive submonoid, containing 0, and well ordered in the usual real order. Put

$$\delta = \min \Gamma_{>0} > 0, \quad \mathcal{H}_\Gamma = \left\{ \sum_{\gamma \in \Gamma} u_\gamma x^\gamma : u_\gamma \in \mathbb{C} \right\}, \quad \mathfrak{m}_\Gamma = \{u : u_0 = 0\}. \quad (2.1)$$

Every subset of Γ is well ordered. We do not require $\Gamma \cap [0, L]$ to be finite.

Lemma 2.1 (Finite additive ancestry). *For every $\rho \in \Gamma$, the set*

$$\mathcal{A}(\rho) = \{\alpha \in \Gamma : \rho - \alpha \in \Gamma\} \quad (2.2)$$

is finite. If $\beta \in \mathcal{A}(\rho)$ and $\alpha \in \mathcal{A}(\beta)$, then $\alpha \in \mathcal{A}(\rho)$. Every ordered decomposition of ρ into positive elements of Γ has length at most $\lfloor \rho/\delta \rfloor$, and there are only finitely many such decompositions.

Proof. If $\mathcal{A}(\rho)$ were infinite, successively taking least unused elements would give a strictly increasing sequence α_n in that set. Then $\rho - \alpha_n$ would be a strictly decreasing sequence in Γ , contrary to well ordering. The hereditary assertion follows from $\rho - \alpha = (\rho - \beta) + (\beta - \alpha) \in \Gamma$. Every positive summand is at least δ , which proves the length bound. For each fixed length, finiteness follows inductively from the two-summand assertion just proved. Only finitely many lengths are possible. $\square$

The same two-set argument proves that the sum of two well-ordered subsets of $\mathbb{R}_{\geq 0}$ is well ordered and has finite coefficient antidiagonals. Consequently, a well-ordered set $E \subset \mathbb{R}_{>0}$ generates a well-ordered monoid: below a fixed real bound only finitely many word lengths occur, and each fixed-length sumset is well ordered. This supplies many admissible examples. The stronger general ordered-group word lemma in [2] is not claimed as a new result here.

Convolution therefore defines a ring on $\mathcal{H}_\Gamma$. Define D strongly by

$$D\left(\sum_{\gamma} u_{\gamma} x^{\gamma}\right) = \sum_{\gamma} \gamma u_{\gamma} x^{\gamma}. \quad (2.3)$$

Finite convolutions give the Leibniz rule. For vector series use $v(y) = \min \text{supp}(y)$, with $v(0) = +\infty$.

2.2 The equation

Fix $A \in \text{Mat}_m(\mathbb{C})$. Consider

$$Dy = Ay + F(x, y), \quad y \in \mathfrak{m}_{\Gamma}^m, \quad (2.4)$$

where

$$F(x, y) = \sum_{\alpha \in \Gamma} \sum_{k \in \mathbb{N}^m} F_{\alpha, k} x^{\alpha} y^k, \quad F_{\alpha, k} \in \mathbb{C}^m, \quad F_{0,0} = 0, \quad F_{0, e_i} = 0. \quad (2.5)$$

Here $y^k = y_1^{k_1} \cdots y_m^{k_m}$ and $|k| = \sum k_i$. The last two conditions are exactly the formal interpretation of $F(0, 0) = 0$ and $F_y(0, 0) = 0$. Constant and linear terms with *positive* x -exponent are allowed.

For fixed γ , a term contributing to $[x^{\gamma}]F(x, y)$ has $|k| \leq \gamma/\delta$. Every x -exponent and every factor exponent in that term belongs to $\mathcal{A}(\gamma)$. Thus the coefficient is a finite sum. If $|k| = 1$, then $\alpha > 0$; if $\alpha = 0$, then $|k| \geq 2$. In either case every solution coefficient used on the right has exponent strictly less than γ .

Lemma 2.2 (Uniform formal valuation gain). *For $y, z \in \mathfrak{m}_{\Gamma}^m$,*

$$v(F(x, y) - F(x, z)) \geq v(y - z) + \delta. \quad (2.6)$$

Proof. Factor the difference of each monomial $y^k - z^k$ into terms with one factor from $y - z$. A positive x -exponent supplies a gain of at least δ . Otherwise $|k| \geq 2$, and at least one additional positive solution factor supplies that gain. Coefficientwise sums are finite, so cancellation can only increase valuation. $\square$

2.3 Absolute Hahn convergence

Choose any norm on $\mathbb{C}^m$; for estimates below take the maximum-coordinate norm and its induced matrix norm. Define

$$\|y\|_r = \sum_{\gamma \in \Gamma} \|y_{\gamma}\| r^{\gamma}, \quad B_r = \ell^1(\Gamma, r^{\gamma}). \quad (2.7)$$

Scalar B_r is a Banach convolution algebra. Write B_r^+ for its positive-support ideal. A formal series is called *convergent* if it belongs to some B_r , $r > 0$.

The analytic assumption on F , imposed from Section 4 onward, is that for some $r_0, s_0 > 0$,

$$\sum_{\alpha, k} \|F_{\alpha, k}\| r_0^\alpha s_0^{|k|} < \infty. \quad (2.8)$$

This is a precise joint majorant condition; it is not inferred from formal support. Put

$$b(r) = \sum_{\alpha} \|F_{\alpha, 0}\| r^\alpha, \quad (2.9)$$

$$L(r, s) = \sum_{\alpha, |k| \geq 1} |k| \|F_{\alpha, k}\| r^\alpha s^{|k|-1}. \quad (2.10)$$

For $s < s_0$ and $r \leq r_0$ these are finite, and

$$b(r) \longrightarrow 0 \quad (r \downarrow 0), \quad L(r, s) \longrightarrow 0 \quad (r, s \downarrow 0). \quad (2.11)$$

The assertions follow by dominated convergence, using the vanished constant-linear coefficients and, for L , a fixed smaller y -radius to absorb the factor $|k|$. For $\|y\|_r, \|z\|_r \leq s$,

$$\|F(x, y) - F(x, z)\|_r \leq L(r, s) \|y - z\|_r, \quad \|F(x, y)\|_r \leq b(r) + L(r, s) \|y\|_r. \quad (2.12)$$

These elementary inequalities supply every analytic fixed-point estimate used below.

3 Finite algebraic parametrization of formal solutions

3.1 Kernel and cokernel splittings

Let

$$\mathcal{R} = \Gamma_{>0} \cap \text{spec}(A), \quad T_\gamma = \gamma I - A. \quad (3.1)$$

Since γ is real, only positive real eigenvalues lying in Γ occur. For $\gamma \notin \mathcal{R}$ put $R_\gamma = T_\gamma^{-1}$ and $Q_\gamma = 0$. For $\rho \in \mathcal{R}$, choose complements

$$\mathbb{C}^m = \ker T_\rho \oplus U_\rho, \quad \mathbb{C}^m = \text{im } T_\rho \oplus W_\rho.$$

Let Q_ρ be projection onto W_ρ , and let R_ρ be the inverse of $T_\rho : U_\rho \rightarrow \text{im } T_\rho$, extended by zero on W_ρ . Then

$$T_\rho R_\rho = I - Q_\rho, \quad P_\rho := I - R_\rho T_\rho \text{ projects onto } \ker T_\rho. \quad (3.2)$$

A kernel and a cokernel need not be the same subspace. This distinction is essential for Jordan blocks.

Choose bases of all $\ker T_\rho$. Let $c \in \mathbb{C}^q$ collect their coordinates, where

$$q = \sum_{\rho \in \mathcal{R}} \dim \ker T_\rho \leq m. \quad (3.3)$$

Denote the corresponding kernel vector at ρ by $C_\rho(c)$, and put $C(x; c) = \sum_{\rho \in \mathcal{R}} C_\rho(c) x^\rho$. Give each parameter c_i the positive weight $\tau_i = \rho$ of its resonance.

3.2 The universal formal candidate

Theorem 3.1 (Formal compatibility is finite algebraic). *For every $c \in \mathbb{C}^q$ there is a unique formal candidate $Y(c) \in \mathfrak{m}_\Gamma^n$ satisfying*

$$Y = C(x; c) + \mathcal{R}F(x, Y), \quad [x^\gamma]\mathcal{R}u = R_\gamma u_\gamma. \quad (3.4)$$

Every coefficient $Y_\gamma(c)$ is a polynomial in c . There is a finite polynomial map

$$\Psi(c) = (Q_\rho[x^\rho]F(x, Y(c)))_{\rho \in \mathcal{R}} \in \bigoplus_{\rho \in \mathcal{R}} W_\rho \quad (3.5)$$

such that $c \mapsto Y(c)$ is a bijection from $\Psi^{-1}(0)$ onto the positive formal solutions of (2.4). Moreover:

1. *Only the finite hereditary set $\mathcal{A}_* := \bigcup_{\rho \in \mathcal{R}} \mathcal{A}(\rho)$ is needed to compute Ψ .*
2. *Only coefficients $F_{\alpha, k}$ with $\alpha \in \mathcal{A}_*$ and $|k| \leq \lfloor R_*/\delta \rfloor$ can be needed, where $R_* = \max \mathcal{R}$ when $\mathcal{R} \neq \emptyset$.*
3. *Each monomial c^ν occurring in Y_γ has $\text{wt}(\nu) := \sum_i \tau_i \nu_i \leq \gamma$.*

For $\mathcal{R} = \emptyset$ the map Ψ is empty and there is one formal solution.

Proof. Define $Y^{(0)} = 0$ and $Y^{(n+1)} = C + \mathcal{R}F(x, Y^{(n)})$. The operator $\mathcal{R}$ cannot lower exponents. Lemma 2.2 therefore gives

$$v(Y^{(n+1)} - Y^{(n)}) \geq (n+1)\delta.$$

Every coefficient eventually stabilizes. Its limit is supported in Γ , hence is a Hahn series, and finite coefficientwise evaluation shows that it satisfies (3.4). If two candidates have a nonzero difference of least exponent β , (2.6) would make the same difference have valuation at least $\beta + \delta$, a contradiction.

Equivalently, at each exponent one computes the finite polynomial

$$f_\gamma(c) = [x^\gamma]F(x, Y_{<\gamma}(c)), \quad Y_\gamma(c) = R_\gamma f_\gamma(c) + C_\gamma(c), \quad (3.6)$$

where $C_\gamma = 0$ off $\mathcal{R}$. The right side uses only lower coefficients. All dependencies lie in the finite set $\mathcal{A}(\gamma)$, which proves polynomiality and finite dependence without making any local-finiteness assumption on the whole low-exponent region.

Multiplying (3.6) by T_γ gives the exact residual

$$DY - AY - F(x, Y) = - \sum_{\rho \in \mathcal{R}} \Psi_\rho(c) x^\rho. \quad (3.7)$$

Thus $\Psi = 0$ is sufficient. For any actual formal solution y , take $C_\rho = P_\rho y_\rho$. Applying R_γ to its coefficient equations gives (3.4), so uniqueness proves necessity and the claimed bijection.

Hereditary ancestry proves the first assertion. At exponents at most R_* , a monomial has at most R_*/δ positive solution factors, which proves the second. Finally, induction in (3.6) shows that the sum of the parameter weights in a product cannot exceed the sum of its solution-factor exponents, which is at most the output exponent. The free term at ρ has exactly its assigned weight ρ . $\square$

Remark 3.2 (A finite certificate is not a bounded ordinary jet). An infinite number of coefficients can lie below R_* . The theorem uses only additive divisors of the *exact* resonances. In particular, “retain everything below the largest resonance” is generally not a finite algorithm. The finite set $\mathcal{A}_*$ can be much smaller.

Proposition 3.3 (Coordinate independence). *Different kernel/cokernel splittings give polynomially isomorphic formal parameter varieties. Their subsets of convergent solutions correspond under the same isomorphism.*

Proof. On a formal solution, the new parameter at ρ is the new kernel projection applied to $Y_\rho(c)$. This is polynomial by Theorem 3.1. Reversing the splittings gives the inverse map, since both parameter maps describe the same solution coefficients. Convergence concerns the solution itself and is unchanged. $\square$

3.3 A sharp degree bound

Treat the finitely relevant entries of $F_{\alpha,k}$ as variables as well as the kernel coordinates c . The entries of A, R_γ, Q_γ are fixed constants.

Proposition 3.4 (Total coefficient degree). *The polynomial Y_γ has total degree at most $2\lfloor \gamma/\delta \rfloor - 1$. The same bound holds for Ψ_ρ with $\gamma = \rho$. This bound is attained for arbitrarily large integer ρ/δ .*

Proof. Expand the recursion into rooted substitution trees. A coefficient vertex $F_{\alpha,k}$ has $|k|$ children and contributes exponent α ; a free-parameter leaf contributes its resonance exponent. A zero-exponent coefficient vertex has at least two children. Every other vertex has positive exponent cost at least δ . The sum of all costs is the output exponent γ .

If P is the number of positive-cost vertices and Z the number of zero-cost vertices, then $P \leq \lfloor \gamma/\delta \rfloor$. Every leaf has positive cost, and a tree has at most its number of leaves minus one vertices of arity at least two. Thus $Z \leq P - 1$ and the number of coefficient or parameter factors is at most $2P - 1$. Finite ancestry and the finite word-length bound make this a finite polynomial expansion, not an unproved infinite tree sum. The sharp family is established next. $\square$

Example 3.5 (Exact sharp obstruction). For $n \geq 2$, consider on $\Gamma = \delta\mathbb{N}$

$$(D - n\delta)y = ax^\delta + by^2. \quad (3.8)$$

The obstruction at $n\delta$ is exactly

$$\Psi_n = \frac{a^n b^{n-1}}{((n-1)!)^2 \delta^{2n-2}}. \quad (3.9)$$

Its total degree is $2n - 1$, proving sharpness.

For completeness, set $z = x^\delta$, $\theta = z d/dz$, and, initially for $b \neq 0$, substitute $y = -(\delta/b)\theta u/u$, $u(0) = 1$. The Riccati equation becomes

$$\theta(\theta - n)u + qzu = 0, \quad q = ab/\delta^2.$$

For $1 \leq j < n$, its coefficients satisfy $u_j = qu_{j-1}/(j(n-j))$, so $u_{n-1} = q^{n-1}/((n-1)!)^2$. The obstruction in the y equation is au_{n-1} , by multiplying the linear residual by $-\delta^2/(bu)$ and comparing at the first obstructed exponent. This is (3.9). Both sides are polynomial in a, b , so the identity extends to $b = 0$.

3.4 Choosing the monoid from the equation

The fixed monoid is not necessarily an artificial restriction. Suppose all coefficient exponents of F lie in a given well-ordered positive monoid Γ_0 . Adjoin all positive real eigenvalues of A and take the generated monoid Γ . It is again well ordered. Every positive real Hahn solution of the equation then lies in $\mathcal{H}_\Gamma^m$.

Indeed, read its finitely many kernel coefficients at the eigenvalues, and construct the candidate in $\mathcal{H}_\Gamma^m$. Both satisfy the coefficientwise fixed-point equation with those kernel data. Their two supports and the input monoid have a common positive lower bound. Factoring their difference as in Lemma 2.2 rules out a least nonzero coefficient of the difference. Thus the arbitrary positive solution equals the constructed candidate. This argument requires positive real support, not a general higher-rank transseries support.

4 All divergence is confined to a bounded exponent window

Assume (2.8). Define the spectral cutoff

$$H = \max(\{0\} \cup (\text{spec}(A) \cap \mathbb{R}_{>0})). \quad (4.1)$$

It includes positive real eigenvalues whether or not they belong to Γ . Write $J = \Gamma \cap (0, H]$. This set may be infinite.

Lemma 4.1 (The tail inverse is bounded). *There is a finite constant*

$$M_H = \sup_{\gamma \in \Gamma, \gamma > H} \|(\gamma I - A)^{-1}\| < \infty. \quad (4.2)$$

Proof. The nonempty set $\Gamma \cap (H, \infty)$ has a least element $\gamma_+ > H$. There is no real eigenvalue of A in $[\gamma_+, \infty)$. The resolvent is continuous and bounded on each compact subinterval of that ray, and for $\gamma > \|A\|$ the geometric-series estimate gives $\|(\gamma I - A)^{-1}\| \leq (\gamma - \|A\|)^{-1}$. Together these estimates prove the claim. The right-hand gap at H follows from well ordering; no left-hand gap has been assumed. $\square$

For an explicitly certified norm bound one may instead use a larger cutoff $H' > \|A\|$, giving $M_{H'} \leq (H' - \|A\|)^{-1}$. The smaller cutoff (4.1) is useful for sharp degree bounds.

Theorem 4.2 (Bounded-window convergence criterion). *For every formal candidate $Y(c)$ of Theorem 3.1, whether or not $\Psi(c) = 0$,*

$$Y(c) \in B_r \text{ for some } r > 0 \iff \sum_{\gamma \in J} \|Y_\gamma(c)\| < \infty. \quad (4.3)$$

In particular, a compatible formal solution cannot acquire a new absolute convergence obstruction above the largest positive real eigenvalue.

Proof. On a bounded interval of exponents, the positive weights r^γ are bounded above and below by positive constants. Thus membership of the truncation in any B_r is equivalent to the right side of (4.3). Necessity follows at once.

For sufficiency put $p = \sum_{\gamma \in J} Y_\gamma(c)x^\gamma$ and write $Y = p + z$. The assumed coefficient sum is finite. For $r \leq 1$,

$$\|p\|_r \leq r^\delta \sum_{\gamma \in J} \|Y_\gamma(c)\|,$$

so $\|p\|_r \rightarrow 0$. Let $\Pi_{>H}$ denote exponent truncation, and let $\mathcal{R}_{>H}$ apply $(\gamma I - A)^{-1}$ on that tail. The required tail fixed-point equation is

$$z = \mathcal{R}_{>H} \Pi_{>H} F(x, p + z). \quad (4.4)$$

Both truncation and the inverse are bounded on B_r , with norms at most 1 and M_H .

Choose $0 < \eta < s_0$. By first making η small and then r small, we can arrange

$$\|p\|_r \leq \eta/4, \quad M_H b(r) \leq \eta/8, \quad M_H L(r, \eta) \leq 1/2. \quad (4.5)$$

The map on the right of (4.4) sends the closed tail ball $\|z\|_r \leq \eta/2$ into itself: at $z = 0$ its norm is at most $\eta/8 + \eta/8 = \eta/4$, and its Lipschitz constant is at most $1/2$. Banach’s contraction theorem gives a convergent tail.

Its formal coefficients equal those of the original candidate. Above H there is no resonance and both tails satisfy (4.4). If they differed, their least different exponent would gain at least δ under the right side by Lemma 2.2, which is impossible. The finite possible formal residuals below H do not enter this argument. $\square$

Corollary 4.3 (Locally finite exponent supports). *If $\Gamma \cap [0, L]$ is finite for every L , every positive formal solution of (2.4) is convergent. In particular this holds for monoids generated by finitely many positive real numbers.*

Proof. The set J is then finite. A finitely generated positive real monoid is locally finite because each generator count is bounded under a fixed exponent cutoff. $\square$

4.1 Analytic evaluation

A convergent Hahn series defines a holomorphic function on the logarithmic cover of $0 < |x| < r$. For $0 < r' < r$,

$$\|Dy\|_{r'} \leq \frac{\|y\|_r}{e \log(r/r')}, \quad (4.6)$$

since $\sup_{t \geq 0} t(r'/r)^t = 1/(e \log(r/r'))$. Thus termwise differentiation is valid on smaller radii. Absolute convolution and (2.8) also justify substitution into F . A compatible convergent candidate consequently solves the actual differential equation on that cover. These statements do not require F or y to extend holomorphically across $x = 0$ in the ordinary sense.

5 The algebraic convergence-locus theorem

This is the central result. The bounded exponent window can contain infinitely many non-summable coefficients. Nevertheless its dependence on finitely many positive-weight resonant parameters has bounded degree. That finite polynomial complexity is sufficient.

Theorem 5.1 (Algebraicity at zero spectral gap). *Assume (2.8). There are finitely many polynomials $\chi_1, \dots, \chi_s \in \mathbb{C}[c_1, \dots, c_q]$ such that*

$$\{c : Y(c) \text{ is convergent}\} = \{c : \chi(c) = 0\}. \quad (5.1)$$

Consequently the positive convergent solutions are parametrized by the affine algebraic set $\mathcal{V}_{\text{an}} = V(\Psi, \chi)$. If $H > 0$, the polynomials can be chosen to use only monomials satisfying

$$\sum_{i=1}^q \tau_i \nu_i < H. \quad (5.2)$$

The number s is at most the number of such monomials. If $q > 0$, their ordinary degrees are at most $\lceil H / \min_i \tau_i \rceil - 1$. If $H = 0$, there is no analytic obstruction.

Proof. Suppose first that $H > 0$. Theorem 3.1 gives, on the entire bounded window J , a uniform finite monomial expansion

$$Y_\gamma(c) = \sum_{\nu: \text{wt}(\nu) \leq H} a_{\gamma, \nu} c^\nu, \quad a_\nu = (a_{\gamma, \nu})_{\gamma \in J}. \quad (5.3)$$

There are only finitely many allowed ν , since all parameter weights are positive. For $q = 0$ the one possible monomial is the constant 1.

Let $\mathcal{E} = (\mathbb{C}^m)^J$ be the vector space of all coefficient families on J , and let $\mathcal{L} = \ell^1(J; \mathbb{C}^m)$ be its linear subspace of absolutely summable families. Form the *algebraic* quotient $\mathcal{E}/\mathcal{L}$; no norm or Hausdorff topology on this quotient is used. The classes $[a_\nu]$ span a finite-dimensional subspace W . Choose a basis $w_1, \dots, w_s$ of W and write

$$[a_\nu] = \sum_{j=1}^s q_{j\nu} w_j.$$

Define

$$\chi_j(c) = \sum_{\nu} q_{j\nu} c^\nu. \quad (5.4)$$

The family $(Y_\gamma(c))_{\gamma \in J}$ lies in $\mathcal{L}$ exactly when its quotient class vanishes, and by linear independence this is exactly $\chi_1(c) = \dots = \chi_s(c) = 0$. Theorem 4.2 proves (5.1).

If $\text{wt}(\nu) = H$, then $a_{\gamma, \nu} = 0$ for $\gamma < H$ by weighted degree. Its only possible nonzero entry in J is at $\gamma = H$. Such a family is summable, so $[a_\nu] = 0$. Those monomials may therefore be omitted, giving the strict inequality (5.2). The equation-count and ordinary-degree bounds follow immediately. When $H = 0$, the set J is empty, and Theorem 4.2 makes every candidate convergent. $\square$

Remark 5.2 (Why this is more than a spectral-gap statement). No lower bound on the nonzero denominators below H was used. Infinite families of unbounded coefficients may cancel after specializing c . The quotient keeps precisely the non- ℓ^1 parts and does not replace such cancellation by an absolute tree majorant. This is why the theorem can describe exceptional convergent parameters in a zero-gap system.

Corollary 5.3 (The top resonant parameters are unrestricted analytically). *A parameter of weight exactly H does not occur in χ . In particular, if all resonant parameters have weight H , the candidates are either all convergent or all divergent. This is a statement about the convergence condition; the separate compatibility equations $\Psi = 0$ still apply.*

Theorem 5.4 (Uniform convergence on compact parameter sets). *For every compact subset $K \subset \{\chi = 0\}$ there is $r_K > 0$ such that $Y(c) \in B_{r_K}$ for all $c \in K$, with uniformly bounded norms. Locally on the algebraic locus $\{\chi = 0\}$, this family is the restriction of a holomorphic Banach-valued map. The assertion remains valid on singular or reducible loci and, in particular, on $V(\Psi, \chi)$.*

Proof. Choose actual coefficient families $\tilde{w}_j \in \mathcal{E}$ representing the basis classes w_j used above. Then

$$b_\nu = a_\nu - \sum_j q_{j\nu} \tilde{w}_j \in \mathcal{L}.$$

For c satisfying $\chi(c) = 0$, its window family equals

$$(Y_\gamma(c))_{\gamma \in J} = \sum_{\nu} b_\nu c^\nu. \quad (5.5)$$

The right side is a finite polynomial with values in ℓ^1 . It is uniformly bounded for $c \in K$. As a positive series its B_r norm tends to zero uniformly on K as $r \downarrow 0$.

The three inequalities (4.5) can therefore be imposed with a common r . The resulting contraction supplies a common tail ball and a common contraction constant. For the holomorphic assertion, use (5.5) as a polynomial definition of $p(c)$ also *off* the locus. On a sufficiently small open parameter neighborhood it obeys the same strict bounds. Iterating the tail map gives holomorphic Banach-valued functions converging uniformly on smaller neighborhoods, so its fixed point is holomorphic. On the locus, uniqueness identifies it with the original formal candidate. Off the locus this extension need not solve the original full equation, and no such claim is needed. $\square$

5.1 What is and is not effective

The formal map Ψ has finite input dependence. The map χ has finite output dimension and bounded polynomial degree, but its coefficients encode which finite linear combinations of the families a_ν lie in ℓ^1 . These summability relations need not be decidable from an arbitrary presentation of the input.

In particular, $\mathcal{E}/\mathcal{L}$ above is not being treated as a Banach quotient, and no continuity of quotient-coordinate functionals is assumed. Choosing a basis of its finite-dimensional algebraic subspace is legitimate linear algebra. Replacing that choice by an effective exact algorithm is a separate question with additional hypotheses.

6 A sharp criterion for universal convergence

Algebraicity describes a fixed equation even when some solutions diverge. There is also a precise condition under which every admissible analytic input has only convergent positive formal solutions. Here the relevant numbers are eigenvalues of A , not the eigenvalue *differences* in a matrix-gauge problem.

Define

$$g(A, \Gamma) = \inf\{|\gamma - \lambda| : \gamma \in \Gamma_{>0}, \lambda \in \text{spec}(A), \gamma \neq \lambda\}. \quad (6.1)$$

Theorem 6.1 (Sharp universal convergence criterion). *For fixed A and Γ , the following are equivalent:*

1. $g(A, \Gamma) > 0$.
2. No positive real eigenvalue of A is a left accumulation point of Γ .
3. For every F satisfying (2.5) and (2.8), every positive formal solution in $\mathcal{H}_\Gamma^m$ is convergent.

If these conditions fail, an F with a genuinely quadratic nonlinear term and a divergent positive formal solution can be chosen arbitrarily small in any prescribed joint norm of the form (2.8).

Proof. Since $\Gamma_{>0}$ has least element δ , zero and negative real eigenvalues cannot be approached. A nonreal eigenvalue has distance at least the absolute value of its imaginary part from real exponents. Only finitely many eigenvalues occur. Unbounded exponents cannot make the distances in (6.1) tend to zero. Finally, no real number can be a right accumulation point of a

well-ordered real set: the part of that set strictly above the number, if nonempty, has a least member. It follows that failure of a positive gap is exactly left accumulation at a positive real eigenvalue. This proves the first equivalence.

Assume $g(A, \Gamma) > 0$. The matrices R_γ in Section 3 then have a common finite norm bound M . For nonresonant exponents this follows from boundedness of the matrix resolvent on bounded closed sets separated from its poles and its decay at infinity. At the finitely many exact resonances the chosen matrices R_ρ are fixed and finite. For any fixed c , the kernel polynomial $C(x; c)$ tends to zero in B_r as $r \downarrow 0$. The estimates (2.11)–(2.12) make $y \mapsto C + \mathcal{R}F(x, y)$ a contraction on a sufficiently small B_r ball after shrinking r . Its fixed point equals $Y(c)$ by formal uniqueness. Thus all candidates, and hence all compatible solutions, converge.

For the converse let $\rho > 0$ be a real eigenvalue approached from below. Select strictly increasing $\alpha_n \in \Gamma$ with

$$\rho/2 < \alpha_n < \rho, \quad 0 < \rho - \alpha_n < 2^{-2n}. \quad (6.2)$$

Choose an eigenvector $v \neq 0$ with $Av = \rho v$ and a linear functional ℓ with $\ell(v) = 1$; ℓ need not be a left eigenvector. For positive constants ε, κ , put

$$F(x, y) = \varepsilon \sum_{n \geq 1} 2^{-n} x^{\alpha_n} v + \kappa \ell(y)^2 v. \quad (6.3)$$

This satisfies the analytic majorant condition at every fixed positive x -radius and finite y -radius. Taking ε, κ small makes its norm arbitrarily small while keeping its nonlinear term nonzero.

Seek a solution $y = uv$. The scalar equation is

$$(D - \rho)u = \varepsilon \sum_{n \geq 1} 2^{-n} x^{\alpha_n} + \kappa u^2.$$

Set its optional free coefficient at ρ to zero. Below ρ , all nonzero forcing exponents exceed $\rho/2$, so the quadratic term contributes nothing. At ρ it contributes nothing either, and the forcing has no term there. Hence a formal solution exists, including when $\rho \notin \Gamma$ so no compatibility equation is present. Its coefficients are

$$u_{\alpha_n} = \frac{\varepsilon 2^{-n}}{\alpha_n - \rho}, \quad |u_{\alpha_n}| > \varepsilon 2^n.$$

Because the exponents stay bounded and tend to ρ , these coefficients are not summable at any positive radius. The full vector uv is an actual formal solution of the matrix equation, regardless of the other eigenvalues of A . This disproves the universal convergence assertion. $\square$

Remark 6.2 (Necessity is a universal, not pointwise, claim). Zero gap does not imply that every equation, or every parameter choice, diverges. The algebraic convergence locus may be empty, all of parameter space, or a proper algebraic subset. The theorem constructs a bad input when the gap fails; it does not replace the cancellation-sensitive criterion for a particular input.

7 Universality: every affine algebraic set can occur

The algebraicity theorem gives a substantial restriction, but it cannot be strengthened to linearity or smoothness. The following construction also separates formal solvability from convergence particularly cleanly.

Theorem 7.1 (Universality of the analytic selection locus). *Let $P_1, \dots, P_s \in \mathbb{C}[c_1, \dots, c_n]$ be arbitrary polynomials. There is a triangular system of the form (2.4), with convergent Hahn*

coefficients and no formal compatibility obstruction, whose formal solution parameter space is $\mathbb{C}^{n+s}$ and whose convergence locus is

$$V(P_1, \dots, P_s) \times \mathbb{C}^s. \quad (7.1)$$

After normalizing the s auxiliary resonant coefficients to zero, the formal parameter space is $\mathbb{C}^n$ and the convergence locus is exactly $V(P_1, \dots, P_s)$.

Proof. Choose $h > 0$ and an integer $d \geq \max(2, \deg P_1, \dots, \deg P_s)$. Set $\rho = (d+1)h$ and

$$g(x) = \sum_{r=2}^{\infty} \frac{x^{h-h/r}}{r^2}, \quad \Gamma = \langle h, h - h/r \ (r \geq 2) \rangle. \quad (7.2)$$

The generating set is positive and well ordered, with least element $h/2$, so Γ is admissible. The series g is absolutely convergent at every positive radius, because its exponents lie in $[h/2, h)$ and $\sum r^{-2} < \infty$.

Write $P_j(c) = \sum_k p_{j,k} c^k$. Use variables $u_1, \dots, u_n$ and $v_1, \dots, v_s$ and equations

$$Du_i = hu_i, \quad 1 \leq i \leq n, \quad (7.3)$$

$$Dv_j = \rho v_j + g(x) \sum_k p_{j,k} x^{(d-|k|)h} u^k, \quad 1 \leq j \leq s. \quad (7.4)$$

All x -exponents in the nonlinear part are positive, and $d - |k| \geq 0$. Thus the structural conditions and the analytic majorant condition hold. The residue matrix is diagonal with entries h and ρ .

Every formal solution of (7.3) is $u_i = c_i x^h$. Substitution in (7.4) gives

$$(D - \rho)v_j = P_j(c) \sum_{r=2}^{\infty} \frac{x^{\rho-h/r}}{r^2}.$$

There is no forcing coefficient at the exact resonance ρ , so every c is formally allowed, and each v_j has an independent free coefficient z_j at ρ . The full formal solutions are exactly

$$u_i = c_i x^h, \quad v_j = z_j x^\rho - \frac{P_j(c)}{h} \sum_{r=2}^{\infty} \frac{x^{\rho-h/r}}{r}. \quad (7.5)$$

At any positive radius, the terms $r_0^{\rho-h/r}$ tend to $r_0^\rho > 0$. The last series is therefore absolutely convergent exactly when its scalar prefactor vanishes. This proves (7.1). Normalizing $z_j = 0$ removes the auxiliary affine factor and changes no convergence condition. $\square$

The ordinary-degree bound in Theorem 5.1 is also sharp. Take one parameter of weight h and a polynomial of exact degree d . Here $H = (d+1)h$, so the bound is d , and the construction realizes a degree- d convergence equation. Choosing a polynomial with d distinct roots ensures that no lower-degree equation defines the same locus in that parameter line.

Example 7.2 (A two-point convergence locus). Take $n = s = 1$, $h = 1$, $d = 2$, and $P(c) = c^2 - 1$. The system is

$$Du = u, \quad Dv = 3v + g(x)(u^2 - x^2), \quad g(x) = \sum_{r=2}^{\infty} r^{-2} x^{1-1/r}. \quad (7.6)$$

All $(c, z) \in \mathbb{C}^2$ give formal solutions,

$$u = cx, \quad v = zx^3 - (c^2 - 1) \sum_{r=2}^{\infty} r^{-1} x^{3-1/r}.$$

The convergent locus is exactly $\{c = 1\} \times \mathbb{C} \cup \{c = -1\} \times \mathbb{C}$. Every coefficient of the equation is convergent; convergence of the solution is selected by nonlinear cancellation, not by an input summability failure.

Example 7.3 (Singular and reducible convergence loci). Using $P(c_1, c_2) = c_1 c_2$ gives the union of the two coordinate axes. Using $P(c_1, c_2) = c_2^2 - c_1^3$ gives a cusp. These examples show that the word “algebraic” in Theorem 5.1 does not hide an unproved regularity assertion. The theorem concerns the complex point set; no claim about a canonical nonreduced scheme structure is intended.

7.1 Formal compatibility can also encode arbitrary varieties

For comparison, omit the accumulating multiplier g and use

$$Du_i = hu_i, \quad Dv_j = dhv_j + \sum_k p_{j,k} x^{(d-|k|)h} u^k. \quad (7.7)$$

Now $u = cx^h$ gives exact-resonance forcing $x^{dh} P_j(c)$. The formal compatibility equations are $P_j(c) = 0$, and the formal solution space is $V(P) \times \mathbb{C}^s$. All those solutions converge, by local finiteness. Thus the same algebraic set can arise either as a formal resonance obstruction or as a purely analytic selection condition. The accumulating construction proves that the two mechanisms are not the same.

8 Nonlinear generation of a sharp small-divisor threshold

The universal gap criterion uses the whole exponent monoid. It cannot in general be replaced by separation of the original forcing support. The following scalar example gives exact formulas and a complete threshold, including compatibility at the resonance.

Let

$$\alpha_n = 2 - \frac{1}{n} \quad (n \geq 3), \quad \Gamma = \langle 1, \alpha_3, \alpha_4, \dots \rangle. \quad (8.1)$$

The least positive exponent is 1, and the only members of Γ up to 3 are

$$0, 1, 2, 3, \quad \alpha_n, \quad 1 + \alpha_n \quad (n \geq 3). \quad (8.2)$$

Indeed, two α -generators have sum at least $10/3 > 3$. Below 3, an integer plus an α -generator can use at most one 1. In particular,

$$\mathcal{A}(3) = \{0, 1, 2, 3\}, \quad (8.3)$$

although infinitely many exponents lie below 3.

For $p > 1$ and $a, b \in \mathbb{C}$ consider

$$(D - 3)y = ax + \sum_{n=3}^{\infty} n^{-p} x^{\alpha_n} + by^2 + 2b^2 y^3. \quad (8.4)$$

Its forcing is absolutely convergent at every positive radius. Each original nonzero forcing exponent is at distance strictly greater than 1 from the spectral value 3; their infimum distance is 1. There is no original forcing at the resonance.

Theorem 8.1 (Exact nonlinear threshold). *Every a, b and every free coefficient $c = y_3$ produce a unique positive formal solution of (8.4). Its coefficients in the critical window are*

$$y_1 = -\frac{a}{2}, \quad y_2 = -\frac{a^2 b}{4}, \quad (8.5)$$

$$y_{\alpha_n} = -\frac{n^{1-p}}{n+1}, \quad y_{1+\alpha_n} = -\frac{ab n^{2-p}}{n+1}, \quad (8.6)$$

$$y_3 = c. \quad (8.7)$$

The formal solution is convergent if and only if

$$p > 2 \quad \text{or} \quad ab = 0. \quad (8.8)$$

The criterion is independent of the top resonant parameter c .

Proof. At exponents 1 and 2, coefficient comparison gives $y_1 = -a/2$ and $y_2 = -by_1^2 = -a^2 b/4$. The exponent α_n is below 2, so no nonlinear term contributes:

$$y_{\alpha_n} = \frac{n^{-p}}{\alpha_n - 3} = -\frac{n^{1-p}}{n+1}.$$

At $1 + \alpha_n = 3 - 1/n$, the only nonlinear contribution is $2by_1 y_{\alpha_n}$. The cubic term starts at exponent 3. Dividing by $-1/n$ gives the second coefficient in (8.6).

At the exact resonance 3, only the ancestors in (8.3) can enter. The forcing is

$$2by_1 y_2 + 2b^2 y_1^3 = \frac{a^3 b^2}{4} - \frac{a^3 b^2}{4} = 0.$$

Hence compatibility holds for every a, b and $c = y_3$ is free. Theorem 3.1 supplies the unique formal continuation.

The first infinite coefficient family in (8.6) is summable for every $p > 1$. The second has magnitude comparable to $|ab|n^{1-p}$. It is summable exactly when $ab = 0$ or $p > 2$. All exponents in this calculation are bounded by 3, so no radius can alter this summability test. Theorem 4.2, with $H = 3$, proves that this is also sufficient for the full solution to converge. $\square$

Remark 8.2 (The two roles of cancellation). The cubic term is chosen to cancel the *exact* resonance, making all formal branches exist. It does not cancel the infinitely many quadratic near-resonances below 3. Thus formal compatibility alone cannot detect the threshold. Conversely, setting $a = 0$ or $b = 0$ removes the entire non-summable family, illustrating why an absolute path bound can miss a convergent specialization.

8.1 Finite coefficient data cannot determine the analytic locus

Proposition 8.3 (No general finite-coefficient convergence certificate). *For a suitable fixed A and Γ , and any prescribed finite set of input coefficient positions, there exist two analytic inputs agreeing at all those positions and having identical formal compatibility maps, while one has convergent positive formal solutions and the other has none. The inputs can be arbitrarily close in the majorant norm.*

Proof. Take the scalar $A = 1$ and $\Gamma = \langle 1, 1 - 1/n \ (n \geq 2) \rangle$. Use $F^{(0)} = 0$ and, after discarding a sufficiently long finite prefix,

$$F^{(1)}(x, y) = \varepsilon \sum_{n > N} n^{-2} x^{1-1/n}.$$

Choose N so that none of the displayed coefficient positions is among the finitely prescribed ones. Both inputs have zero forcing at the exact resonance 1, hence their compatibility maps are identically zero. The first has solutions cx . The second has solutions

$$cx - \varepsilon \sum_{n>N} n^{-1} x^{1-1/n},$$

none of which is absolutely convergent. The norm of the difference tends to zero with ε , or with $N \rightarrow \infty$ at fixed radius. $\square$

This does not contradict the finite polynomial description of the analytic locus. In the example the extra polynomial equation is simply a nonzero constant, so the locus is empty. Determining that constant obstruction requires information about an infinite coefficient tail.

9 Computation, proof audit, and a formalization route

9.1 What the supplied program checks

The program `verification/verify.py` uses exact rational exponents, exact SymPy matrices, and symbolic polynomial coefficients. Its central routine enumerates a *finitely generated rational* monoid up to a cutoff and implements (3.6). At a singular T_ρ , it constructs a right inverse by pivot columns and rows, rather than assuming that kernel and cokernel projections coincide.

After constructing each candidate, the program independently reevaluates F and checks the residual identity (3.7) at every enumerated exponent. Thus an incompatible candidate is expected to have exactly the recorded resonant residual, not a spuriously vanishing one.

The recorded run used Python 3.13.5 and SymPy 1.14.0, with seed 20260929. It produced the following actual counts:

Check category	Count
Finite nonlinear systems	21
Exact scalar equalities, including matrix entries	670
Sharp Riccati coefficient identities in the independent recurrence	24
Seeded additional matrix systems	12

The last two rows are included in, not added to, the equality and system counts. The systems include six sharp-degree examples, the generated small-divisor example, the arbitrary-variety construction, a Jordan-block example, and twelve seeded systems.

For the sharpness family, the first coefficients of the obstruction after removing $a^n b^{n-1} \delta^{-2n+2}$ are

n	Coefficient	Total degree
2	1	3
3	1/4	5
4	1/36	7
5	1/576	9
6	1/14400	11
7	1/518400	13

These agree exactly with (3.9).

9.2 A Jordan-block check

Take

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad F = x \begin{pmatrix} b \\ a \end{pmatrix}.$$

At exponent 1, T_1 has image spanned by the first coordinate and kernel also spanned by the first coordinate, but the cokernel complement is the second coordinate. The correct compatibility condition is $a = 0$, and the candidate coefficient is $(c, -b)^\top$. Projecting onto the kernel in place of the cokernel would give the wrong condition. This exact test exercises a distinction on which the general theorem depends.

9.3 Limits of the computation

The code does not prove that an arbitrary infinite set is well ordered, that an infinite coefficient family is summable, or that the algebraic quotient construction is effective. Restricting an infinite example to finitely many generators verifies its displayed coefficient identities, not its convergence threshold. The threshold and universality arguments are mathematical proofs in Sections 7 and 8.

No Lean theorem is supplied or claimed for the new results. The repository’s `NeumannWords.lean` was inspected as an available support-theory component, not recompiled in this work. The numerical and symbolic checks must not be described as a proof-assistant verification.

9.4 A useful implementation contract

For effective finite data, the core formal algorithm is:

1. Determine the exact real resonances and certify their finite ancestor sets. Choose kernel and cokernel splittings over the coefficient field.
2. Process only those ancestor exponents needed by the requested certificate. Compute f_γ , then $R_\gamma f_\gamma + C_\gamma$, and record $Q_\rho f_\rho$ at a resonance.
3. For convergence, separately represent the window coefficient families and certify their linear relations modulo ℓ^1 . Only after that step construct the analytic equations χ .

The third step cannot generally be replaced by a finite coefficient cutoff, as Proposition 8.3 shows. An infinite-support API must expose exact ancestry and summability certificates, rather than pretending that ordinary bounded enumeration always terminates.

9.5 Formalization in layers

A Lean development can keep the difficult parts separate. The support layer can reuse positive-word and antidiagonal lemmas already present in ProveIt. The finite-dimensional layer needs right inverses, kernel/cokernel splittings, and the coefficient residual identity. The formal layer proves the uniform valuation gain and finite polynomial ancestry.

The analytic layer then builds the weighted ℓ^1 algebra, coefficient projections, resolvent bounds, and the tail contraction theorem. The central algebraicity step is a finite-dimensional linear-algebra argument in the quotient of all window families by the summable subspace. It requires no computability claim and no topology on that quotient. The uniform-parameter theorem additionally needs holomorphic dependence of a uniformly contractive Banach-space fixed point.

A sensible first formal target is Theorem 4.2; it isolates the new analytic reduction and is independent of the choice of quotient basis. The universality construction is a second tractable target because all its coefficients and summability decisions are explicit.

10 Further research questions and concrete next targets

10.1 Effective summability quotients

Theorem 5.1 reduces analytic classification to linear relations among finitely many coefficient families modulo ℓ^1 . For which exact sequence classes is this relation problem decidable? A first target is a family whose coefficients admit certified expansions in $n^\alpha(\log n)^\beta$ with summable remainders. One should prove that cancellation of the finitely many non-summable asymptotic types is both necessary and sufficient, then extract χ algorithmically. Oscillatory families require absolute, not merely conditional, summability.

10.2 Minimal analytic-obstruction dimension

The proof bounds the number of equations by a monomial count, which can be very loose. Determine the possible dimensions of the quotient span W for a prescribed spectrum and coupling pattern. A concrete target is to classify two-component quadratic systems with two positive resonances, including when one obstruction polynomial suffices. The relevant invariant should retain cancellations rather than count all substitution trees.

10.3 Polynomial families of equations

If finitely many external parameters enter the coefficients with uniformly bounded polynomial degree, a bounded exponent window again has a finite polynomial template. Formulate a joint parameter theorem including both input parameters and resonant solution parameters, under a common analytic majorant on compact sets. Sharp degree bounds should account separately for positive exponent cost and zero-cost parameter dependence. Without a polynomial hypothesis, an algebraic conclusion is not expected.

10.4 Moving eigenvalues and wall crossing

Our matrix A is fixed. When its eigenvalues vary, the resonant parameter space changes and the coefficient inverses become rational functions of the spectral parameters. Can one obtain algebraic or constructible convergence loci on a suitable stratification away from moving accumulation walls? The scalar multiplier $(\gamma - \lambda)^{-1}$ already shows why one cannot simply repeat the finite-degree proof with λ treated as another polynomial parameter.

10.5 Logarithmic solutions after compatibility failure

The current theory deliberately excludes logarithms. If $\Psi(c) \neq 0$, which nonlinear systems can be solved after adjoining a bounded degree in $\log x$, and when does the nonlinearity force an unbounded logarithmic expansion? A first test family is the sharp Riccati example (3.8), whose linearization gives explicit resonant behavior. The goal is a theorem distinguishing a finite log correction from a change of asymptotic scale.

10.6 Higher-rank exponent groups

The finite-degree argument uses the real bound $N\delta \leq H$ to bound N . It need not survive in a non-Archimedean ordered group, even though classical Neumann word finiteness remains available for each exact exponent. Find a substitute for a bounded real cost window that still bounds the number of parameter monomials relevant to analytic failure. A useful first result would be either a rank-two positive theorem under a cofinality hypothesis or an explicit nonalgebraic parameter locus.

10.7 Irregular equations and unbounded dangerous exponents

For $x^{k+1}y' = Ay + F(x, y)$ with $k > 0$, coefficient transport is no longer purely diagonal in the exponent. Classical factorial divergence can persist at arbitrarily large exponents, so the bounded-window argument does not apply directly. Determine whether the parameter locus of a fixed Gevrey or Borel-summability class is still algebraic in a carefully chosen irregular model. This is a separate problem, not a consequence of the present theorem.

10.8 Summation beyond absolute convergence

The divergent candidates in the universality construction may still have renormalized analytic realizations. Classify realizations compatible with multiplication and D , and identify which normalization data are invisible to finite asymptotic prefixes. In that enlarged class the selected parameter locus could differ from $\mathcal{V}_{\text{an}}$. Absolute convergence must remain explicit in every comparison.

10.9 Geometry, stability, and certified computation

The convergence locus is algebraic for a fixed equation but can change under arbitrarily small input perturbations. Characterize subclasses in which its dimension or defining ideal is stable. A practical first target is a finite collection of certified non-summable asymptotic types with uniformly summable remainders; the problem may then reduce to finite matrix rank stability. Such a theorem would give meaningful perturbation certificates rather than an unjustified generic robustness claim.

10.10 A complete machine-checked model case

Formalize the two-point system (7.6), including positivity and well ordering of its generated support, exact formal coefficients, and the equivalence of convergence with $c^2 = 1$. This would provide a compact end-to-end test of support algebra, coefficient recursion, absolute summability, and parameter cancellation. The proof should be independent of numerical truncation.

11 Conclusion

The nonlinear problem has two separate algebraic geometries. Exact resonances give a finite-input compatibility variety. Absolute convergence gives another algebraic condition, obtained from possibly infinite coefficient families rather than from a finite input jet.

The decisive analytic fact is that, for positive real Hahn exponents in a regular-singular Euler equation, every possible divergence lies below the largest positive real eigenvalue. Above that cutoff the resolvent is bounded and the nonlinear tail is contractive after shrinking the

radius. Positive parameter weights make the remaining infinite window a finite polynomial family. Passing to the algebraic quotient by ℓ^1 therefore produces finitely many convergence equations.

The universality construction shows both the strength and the limit of this classification: the convergence locus cannot be an arbitrary nonalgebraic subset, but it can be an arbitrary affine algebraic set. The generated small-divisor example further shows that original forcing separation and exact-resonance cancellation do not by themselves imply convergence. These results provide a proof-based extension of the repository’s linear accumulation analysis, with the unresolved effective, logarithmic, and higher-rank questions stated separately.

A Proof dependencies and hypothesis audit

Result	Essential hypotheses and dependency
Formal parametrization	Positive real monoid; well ordering; no constant or zero-exponent linear term in F . Analytic convergence is not used.
Finite ancestry	Exact additive divisors, not finiteness of a bounded interval. The real lower bound supplies an explicit degree bound.
Tail convergence	Joint analytic majorant; fixed finite matrix; positivity of the unknown; well-order right gap above the final real eigenvalue. No left spectral gap is needed.
Algebraic analytic locus	Finite resonant parameter set with positive weights; bounded-window criterion; algebraic quotient by the ℓ^1 subspace. No topological quotient or effective basis is assumed.
Uniform radii	A finite polynomial representation by summable coefficient families on the locus, followed by a common contraction estimate.
Universal gap converse	Quantification over every admissible F ; an invariant eigenvector direction and a selected left-accumulating sequence. The constructed nonlinear formal solution exists.
Universality	A fixed well-ordered accumulating monoid and polynomial coupling in the unknowns; all input coefficients are absolutely summable. Auxiliary free resonant coefficients account for the affine factor.

A.1 Frequently tempting but invalid shortcuts

A bounded set of Hahn exponents need not be finite. A formal coefficient recursion does not imply absolute convergence. An eigenvector is not necessarily paired with a left eigenvector having nonzero pairing, so the universal counterexample uses only an arbitrary functional $\ell(v) = 1$ and explicitly constructs a solution in the invariant line. A kernel projection does not replace a cokernel obstruction for a Jordan block. The quotient by summable families is used algebraically, without an unjustified continuity assertion. Finally, finite algebraicity of a parameter locus does not mean finite-coefficient decidability of that locus.

B Repository provenance and literature boundary

The source comparison used the following pinned commit:

3d5973524506411392a911470b5ddc35521568ea

In particular, the linear Hahn–Fuchsian package was read in its README, introductory framework, explicit accumulation model, verification discussion, and further-research sections. The support module `Algebra/SurrealNumbers/Surreal/HahnSeries/NeumannWords.lean` was inspected through its definitions and principal word-finiteness proof. Repository searches also checked related resonance and convergence-locus terminology. This is not a line-by-line audit of the entire repository or its large canonical transseries volume.

The cited Gontsov–Goryuchkina sources were checked for their stated convergence setting and finite-generation framework. Their results, van der Hoeven’s generalized-series fixed-point theory, and existing Hahn convergence definitions are acknowledged as prior theory, not relabeled as new contributions. The proof of the central algebraic-locus theorem is self-contained, but independence of historical discovery remains a matter for further specialist review.

The delivered package includes the standalone source, compiled PDF, reproduction script, recorded exact results, a source audit, and build instructions. No repository files were modified. No private data or external font files are required to rebuild the article.

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